

A Review of State Art of Text Classification Algorithms

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Abstract— In the recent years, the categorization of text documents into predefined classifications has perceived a growing interest due to the growing of documents in digital form and needs to organize them. Text categorization is one of the extensively used for natural language processing (NLP) applications have achieved using machine learning algorithms. Text classification is a challenging researcher to find the best suitable structure and technique. Classification process done using manual and automatic classification. This research paper covers the preprocessing, feature extraction, different algorithms and techniques for text classification and finally evaluates the performance metrics for assessment.

Keywords—Text Classification; Preprocessing; Feature Extraction; Classification Algorithms; Evaluation Measures; Natural Language Processing; Deep Learning Models;

I. INTRODUCTION

Text classification is the technique of categorizing text or tags into organized classes according to its content. Text classification also called text tagging or text categorization. It is one of the significant tasks in natural language processing (NLP) with extensive applications consisting of intent detection, topic labeling, spam detection, and sentiment analysis. By using NLP, Text classifiers may automatically identify text after which a pre-described collection of tags or categories are allocated according to their topics from medical studies, files, documents and all over the world. Although the classifier defines the category of textual content are classified, the similarity of all inputs in the training sets should be determined. Thus, classification performance assessment will fall down with the rise of training inputs [1]. The classification of the automated arrangement of documents became the main function for knowledge discovery and information organization as the availability of electronic documents and the growth of the internet increased. [2].

Natural language processing (NLP), machine learning techniques and data mining used to identify and discover patterns in electronic documents automatically. Text mining's main purpose is to enable users to extract information from textual tools and to deal with operations. The aim of Information Extraction (IE) methods is to extract precise data from text documents. This is the first approach, which suggests that text mining is the same as data extraction. The process of finding documents that provide answers to

questions known as information retrieval (IR). To achieve this, statistical measurements and methods for automated text data processing and reference to the given query are used. In its largest context, information retrieval encompasses the entire spectrum of information processing, from data retrieval to knowledge retrieval. Text classification contains the steps for processing classification are followed as pre-processing, feature extraction and classification algorithm models for training and prediction as shown in figure1. Training Data set are used for train the data and prediction is used for find the labelled output after classification of text document with different machine learning algorithms as different classifier models.

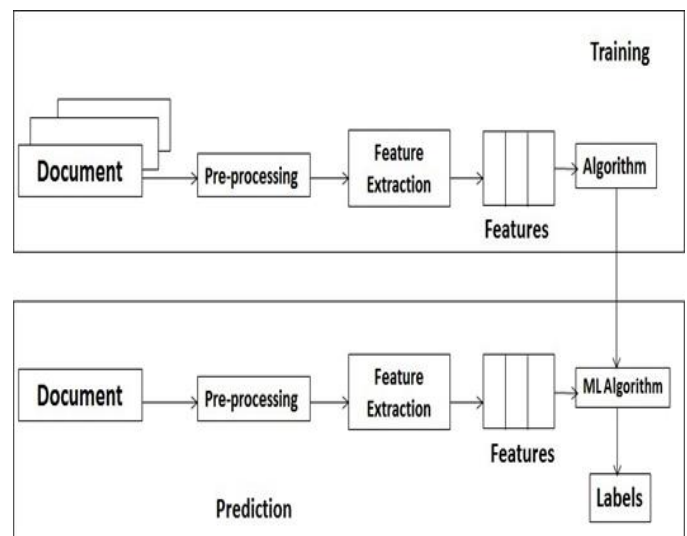


Fig.1. Text classification Process

A. Pre-processing

Pre-processing is the important task for classification of text in natural language processing. There are three main components for processing. Those are Tokenization, Normalization and Noise removal.

Tokenization: Tokenization involves strings that divided into smaller tokens. It is possible to tokenize paragraphs into sentences and tokenize sentences into words.

Normalization: Normalization objects for eliminate common and frequent words like the, a etc. and remove irrelevant words.

Noise removal: Noise removal used to removes the unwanted text and it converts different form of words into similar canonical words [3].

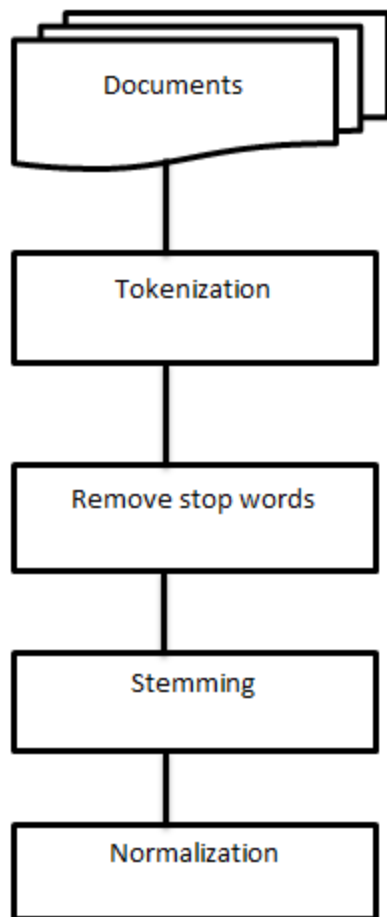


Fig.2. Text classification Preprocessing

B. Feature extraction

The important task of text classification is feature extraction. It gives not only more accuracy and gives greater probability, saves computational time. Feature extractions have different characteristics are word bag and embedding and extract the different characteristics from text [4]. The features converted into different vectors as shown in fig. 1. The vectors are counted in the form of TF-IDF Vectors. TF-IDF is a Term Frequency and Inverse Document Frequency for two-word vectors [5]. For TF, the frequency of occurrence in a text of a word t, d is determined and assigned to the same word as weight. The value of word in the text document calculated by the IDF. By using the following equation, the IDF of word t was determined

$$IDF_t = \log \frac{N}{DF_t} \quad (1)$$

The number of documents shows where N is. Now, using the following equation, the TF-IDF formula for word t are estimated in document d

$$TF-IDF = TF * IDF_t \quad (2)$$

After preprocessing and feature extraction, going to the next step is classify the text documents using Machine learning classification techniques. Based on these algorithms, documents classified to single class or multi class labels. Finally, the performance evaluation measures used for finding the accuracy and recall. Based on these two measures, can identify which classification algorithm performed well and takes less time for computation for classified documents.

II. LITERATURE SURVEY

Compared to the old text classification algorithm, the KNN algorithm has less accuracy and reduces the time complexity, significantly increases the efficacy of text classification for classifying the text. As compared to other machine learning algorithms, Naive Bayes performs well on textual and numeric data, but its conditional independence principle violated by real-life data, it ignores the frequency of word occurrences, and it performs poorly when features are strongly correlated. So, this algorithm used for email classification and spam filtering.

In machine learning algorithms for text classification, SVM classification is the most effective algorithm comparing with all supervised Machine Learning algorithms. SVM for better capture the in-built characteristics of the data. Moreover, the ability to learn can be independent of the function space's dimensionality and global vs. local minima. When it comes to kernel selection and parameter tuning, the SVM has some difficulties. In comparison of other algorithms, KNN and Naïve bayes are suitable for pre-processing. KNN is suitable for continues to achieve good results, in the case of increasing the number of documents, but this is not possible with SVM.

In KNN difficult to find K optimal value and it takes more classification time. In addition, more work is required for improvement of performance and classification of document accuracy. In increased electronic documents, new algorithms and solutions used for knowledge beneficial. Here, rough set method used with KNN text classification algorithm that is improved KNN text classification algorithm. Finding similarity based on distance measures to improve classification performance [6].

This paper presents the importance of different text classification techniques and challenges to solve with these algorithms. It shows about variant techniques in feature selection for finding the better accuracy. This semantic and ontology methods are used for information retrieval and document classification. The paper explains about classification evaluates the performances of the classifiers based on different training text and high or good quality training measures may originate classifiers of good evaluation of performance [7].

Jing Ouyang says that Support Vector Machine (SVM)-based text classification and device data parameters improve when specific experiments conducted, so the classification algorithm significantly improves the efficiency and accuracy of the classifier for text classification [8].

The authors explains about different applications are categorized into topic classification, which is classify text based on topic and relevance or aspect, and sentiment analysis for detecting sentiment in text gives the positive or negative as result and intent classification is classify text based on intent for example feedback, request and complaint [9].

The following parameters used for measuring the performance of classifiers: True positive (TP) as correctly identified, False positive (FP) as incorrectly identified, True negative (TN) as correctly rejected, False negative (FN) as incorrectly rejected [3]. Precision, recall, F-Measure, Micro, and Macro Average used in the text classification evaluation index [10].

These parameters are used for medical data set, in this the algorithm used different modules for classification wither more accuracy. These methods of feature learning module having word embedding layer, positional encoding, attention mechanism, gradient reversal layer and adversarial network for word extraction module based on adversarial network and text classification based on dual channel subtraction [11].

This paper, two distance measure techniques used for better accuracy for bangla text. Different classification algorithms use input text divided into domains such as business, state, science, sports, and medicine. The classification techniques are Euclidean distance, cosine similarity, J48, Naïve bayes, naïve bayes multinomial, random forest and simple logistic. The Euclidean distance and cosine similarity generates best accuracy like 95.20% and 95.80% [13].

This paper uses the TF, IDF, and CHI to find the classification of legal provisions and characteristics words. Here, the class belongs to and does not belongs to classes are defined with A, B, C and D for finding the measures. The English legal text giving the input parameters are society, military, sports, finance, IT, property, education, entertainment are used for traditional TF-IDF and improved TF-IDF algorithms gives better performance [14].

For patent text classification, multivariate neural network fusion used. In this parameters are Model parameter Word embedding dimension, Convolution kernel size set, Number of BIGRU nodes, Dropout, Batch size, Epoch size, Optimizer and Loss having the Parameter values like 300, (3,4,5) , 128 0.5, 64, 100, rmsprop, Categorical crossentropy [15].

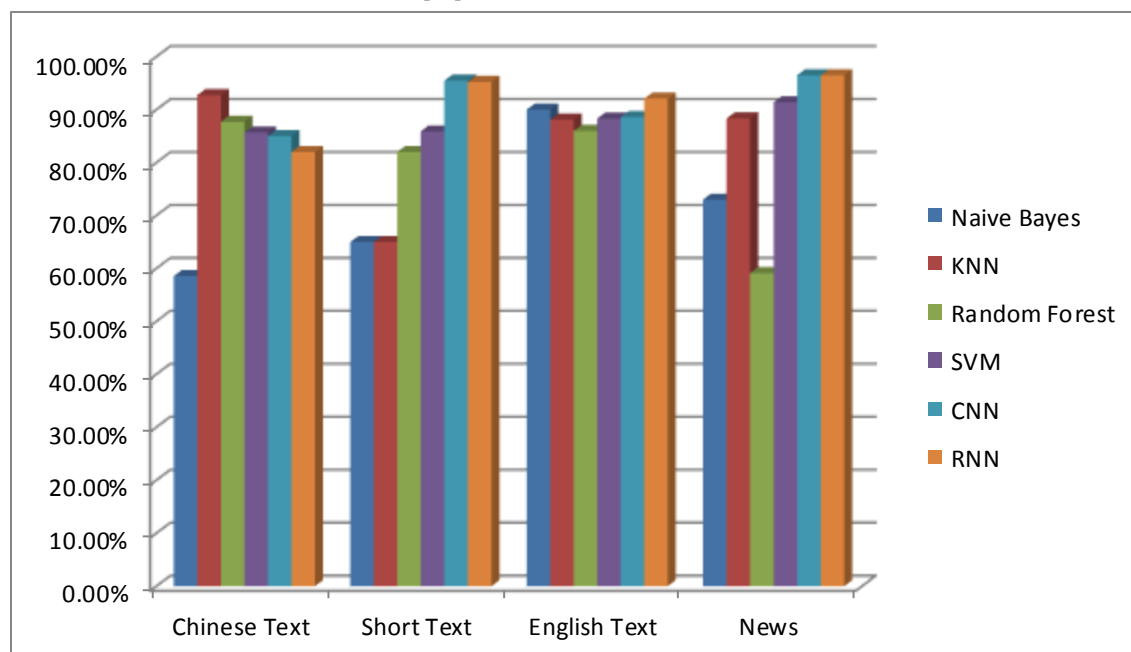


Fig.3. Comparison of accuracy for ML techniques

Unsupervised learning algorithm is clusters used for text classification, removing noisy data, and compared to Bayes algorithms for incremental learning algorithms; the cluster algorithm improves efficiency and improves the accuracy of incremental learning. [12]. For English text, the author used two different SVM algorithms are SVM and improved SVM. Therefore, in this the improved SVM has the following characteristics. Those are good fault tolerance, based on the

user requirement filter the information, stable process running, very few system bugs, speed of the system also good, and takes less time for processing.

In text classification, pre-processing is the first and primary phase. The pre-processing for dataset having the different steps followed by plaintext as input data, data set partition as divided into different word bags. The data vectorization as different n-gram of data, divided into training

set and test set after that build the training dataset for multiple algorithms. Finally predicting the test set based on the result of training set and calculate the accuracy for all algorithms. In this paper, textCNN, TextRNN and TextRNN + Attention methods used for text classification. The text having different forms with special sequence and this context maintain the syntax and semantic dependence. In this paper, the implemented algorithms are deep learning algorithms of conventional neural network (CNN) and recurrent neural network algorithm (RNN).

The RNN technique can deal with semantic dependences to attain significant classification but cannot parallelize well. So this algorithm is suitable for short text. For long text, it takes more training time when the processing text. The CNN can parallelize well for achieving classification. This is suitable for long text and different size of filter of text. This algorithm uses multiple channels and the most influential can select using max pooling. For feature selection, it uses a dropout of full connection layer for getting the result of classification. These algorithms are generated the result of accuracy like CNN gives 85% and RNN gives 82%. Based on Text detection and image detection, we can also do the text classification using convolutional Neural Networks [16].

The classification of text has more advanced, and it requires six key components. The components are data collection, data processing and labelling, extraction and weighting of features, selection or projection of features, model training and evaluation of solutions. For the fast text data the algorithms SVM, KNN, Naïve Bayes had generated an accuracy of 91%, 88% and 86%. [17].

In feature extraction, the word tags divided into unigram and bigram word for easily finding the accuracy and fast classification of text document using TF-IDF. Using British English and American English, the algorithm SVM generate 92.1% of accuracy in stemming test compared with normal text. While unigram text has given accuracy of 91.2% compared with bigram and N-gram, by using linear kernel accuracy 87.1% was generated compared with different kernel algorithms, taking threshold value from 1 to 10 highest frequency of 94.0% was generated at threshold value 2 [18].

The Hadoop Map Reduce facing many limitations in the field of text classification with K-Means. This paper proposed hadoop Map Reduce with Naïve Bayes classifier algorithm for reducing these problems. This classifier is much better for time and space complexity. It is easy to work and simple. In the medical data set, for hadoop map reduce 47.85%, for Gaussian naïve Bayes 72.13%, for Bernoulli naïve Bayes 65.5%, multinomial naïve Bayes 52.45% of accuracy generated. [19].

In this paper, the traditional TF-IDF algorithm cannot return the dissemination of feature words in different categories. Therefore, the author improved this algorithm as improved TF-IDF algorithm and this method implemented with weighting factor, which gives the degree of inter class distribution and intra class distribution, and association between class and feature words. This paper used the Word2vec model for represent the vectors of text word in

place of vector space. This implemented with word vector and weight of word for Word2vec model. Micro average and macro averages used for performance evaluation. The micro average gives more accuracy than macro average for both traditional and improved TF-IDF. The improved TF-IDF gives more accuracy for classification with comparison of traditional TF-IDF and produces the result more effectively [20].

The above authors discussed about different machine learning algorithms and deep learning algorithms. These algorithms are applied for Chinese text, short text, English text and news text and those domains are generated the accuracy as shown as above fig 3. Nowadays, mostly and updated text classification algorithms are deep learning techniques with sub domains and these techniques gives more accuracy and takes less time for computation. Finally, these algorithms are appropriate for large amount of data with multiclass text classification.

III. TEXT CLASSIFICATION ALGORITHMS

Machine learning algorithms classified into unsupervised, supervised, and semi-supervised approaches used to classify documents. Many techniques and algorithms for clustering and classification of electronic documents recently proposed. Using the current literature, this section focused on supervised classification methods, emerging technologies, and some of the opportunities and challenges. As the internet use rate has rapidly increased, the automated sorting of documents into predefined categories has been observed as an active attention. The role of automatic text classification extensively studied in recent years, and rapid progress appears to make in this field, including Machine Learning approaches such as K-Nearest Neighbors (KNN), Naïve Bayes, Support Vector Machine (SVM), Decision Tree, Neural Networks, Convolutional Neural Network (CNN), and Recurrent Neural Network (RNN). Automatic text classification typically employs supervised learning techniques, in which pre-defined category labels assigned to documents based on the probability indicated by a training collection of labelled documents. Some of these methods shown below:

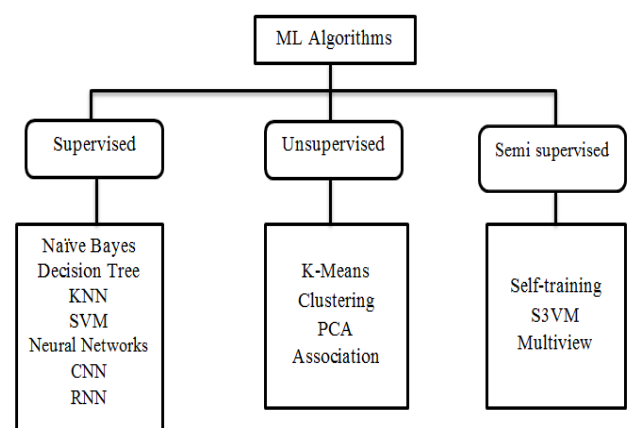


Fig.4. Machine Learning Algorithms

A. Linear Regression

It is the basic and primary method in machine learning for text classification. The linear regression used for finding the estimate real values based on independent variables. For fitting the linear line, we are giving the relationship between dependent variables and independent variables. This best linear line is called as regression line and the equation was represented as $y=mx+c$.

Where y as dependent variable,
x as independent variable,
m as intercept slope and
c intercept constant.

The m and c coefficients calculated by minimizing the number of squared differences in distance between regression line and data points. Linear regression classified into two categories. Those are simple linear regression and multiple linear regressions. Simple linear regression is derived using one independent variable. Multiple linear regressions calculated by using more than one independent variable. We are using curve linear regression or polynomial regression for fitting the linear regression line.

B. K-Nearest Neighbors (K-NN)

KNN is the Machine Learning algorithm for regression and classification problems. The KNN algorithm is a simple method that stores all available cases and classifies new cases based on a majority vote of its k neighbours. The case that assigned to the class is the most common among its K closest neighbours, as determined by a distance function. These distance functions can be Minkowski, Euclidean, Manhattan, and Hamming distance. These three functions are used for continuous function and Hamming for categorical variables. KNN is a nonparametric classification technic for text classification. First, identify the text, which classified for finding the similarity. Next, using KNN to identify the number of neighbor's using K value for text classification. This k value identification is important because it will affect the accuracy for text classification [1]. Since the KNN is computationally costly, variables should be normalised to avoid biasing it. Prior to using kNN for things like outlier detection and noise reduction, spend more time on the pre-processing level.

C. Naïve Bayes

It's a classification method based on Bayes' theorem and the presumption of predictor independence. A Naive Bayes classifier, in simple terms, assumes that the existence of one function in a class is unrelated to the presence of any other feature. Naïve Bayes algorithm used for text classification because of it gives more effectiveness and simplicity. Compare with the simple naïve Bayes and multinomial naïve Bayes have huge advantages and it gives more accuracy.

Naïve Bayes allows one to quantify the characteristics using the conditional probability of two events happening, depending on the probability of each actual occurrence. Therefore, for a given text, we calculate the probability of each word tag. In this theorem, bayes provides the calculation of posterior probability $P(c|x)$ from $P(x), P(c)$ and $P(x|c)$. Bayes_rule equation given as

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)} \quad (3)$$

$$P(c|x) = P(x_1|c) * P(x_2|c) * \dots * P(x_n|c) * P(c)$$

Where,

$P(c|x)$ as the posterior probability of class (target) for the given predictor (attribute).

$P(c)$ as the class prior probability.

$P(x|c)$ as the likelihood, which is the probability of a predictor being assigned to a specific class.

$P(x)$ as the predictor 's prior probability.

D. Support Vector Machine

SVM is one of the best-supervised machine learning algorithms used for regression and classification but mainly used for different problems in classification. In this every data item placed in high dimensional space and after that identify the hyperplanes based on the features. These hyper planes are used for differentiating the classes based on the classification. SVM gives good precision but less in recall for text classification.

The customized SVM improves the recall using different threshold value. It gives the best result for precision and recall for classification. In different ways of classification using SVM improves efficiency and faster during runtime. For high dimensional data, the SVM gives more accuracy because of it controls the complexity and issues are fitted. SVM is a most important text classification machine-learning algorithm with linear transformation. It requires more resources for classification compare with Naïve Bayes, but it gives more accuracy and faster. SVM uses hyper plane for separating tags (vectors) for input features.

E. Decision Tree

For classification problems, we are using the decision tree method is one type in supervised learning algorithms. This technique used for both continuous dependent variables and categorical variables. The population fragmented into two or more similar groups. This achievement based on the most important independent variables/ attributes in order to create as many distinct classes. The best feature for every node of the tree can selected by using attribute selection measure. For this attribute selection measure, we have two techniques; they are Information gain and Gini index.

The decision tree constructs document manual categorizations based on true or false outcomes. It forms the leaf-based tree representing the document type, and branches reflect the combination of features for those categories [2].

F. Neural Networks

For text classification, neural networks have been capable of achieving significant performance in document modelling and sentence. There are two types of neural networks: the Convolutional Neural Network (CNN) and the Recurrent Neural Network (RNN).

CNN filters into different windows with different sizes for features of the documents using layers in CNN. These windows used for classifying the features generating the word tag vectors. The input layer data extract to different features by using a kernel or filter that is convolution. The input data use the dot product of vectors for convolution. These dot product vector displayed result as zero when the text belongs to same class. The embedded word vector used for finding the similarity between vectors. This filter used to count the number of filters in the convolution, and these are depending on the user's requirement. After convolution, the filter generates the multiple features, and these steps are coming under feature extraction of input data based on the layers. After generation of features use the activation function for those filters and it displays the output with nonlinear relationship over the output. These convolutions dimensionally reduced to reduce the complexity and computational time by using activation function. CNN can improve by adding input layers used for more dense layers and a smaller number of wide layers.

RNN is a sequence model to deal with variable length document sequence and long-term dependencies. These two NNs are using the long short-term memory network (LSTM). This LSTM to extract higher-level sequences of tag features.

IV. PERFORMANCE EVALUATION MEASURES

The Text Classification Evaluation Measure is determined according to accuracy, recall, and F-measure. Using a single recall and accuracy to test the classifier is irrational. The evaluation measures are depending on the classification classes. The document belongs to class is dependent on some parameters. Those parameters are true positive (TP), false positive (FP), true negative (TN) and false negative (FN). Using these parameters, the evaluation measures calculated as shown below:

$$\text{Accuracy} = \frac{TP}{TP+FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (5)$$

$$\text{F-measure} = \frac{2 \cdot \text{Accuracy} \cdot \text{recall}}{\text{Accuracy} + \text{recall}} \quad (6)$$

The multi-class text classification technique was use the micro average and macro average for classification evaluation measure. The macro average is the mean of all classes F-measure values.

$$\text{Micro average} = \frac{TP+NP}{TP+TN+FP+FN} \quad (7)$$

The efficiency of the classifier as measured by the above parameters is as follows:

Table 1. Classification evaluation matrix

	Belongs to class	Does not belongs to class
Classification belongs to group	TP/Correctly identified	FP/ Incorrectly identified
Classification does not belongs to group	FN/ Incorrectly rejected	TN/ correctly rejected

V. CONCLUSION

Text have major role in the present era because of information is unstructured and different types of data. Many issues confronted with text existence, it is difficult and time-consuming to understand, interpret, sort and organize through text data. Text classification algorithms used for classifying the text in different applications like legal documents, emails, social media, surveys, chatbots and more in a quick and efficient way. This paper reviewed about text classification using different machine learning algorithms as classifier techniques. Comparison of different machine learning algorithms, SVM gives more accuracy and less time for computation. Decision tree is used for hierarchical type of classification and KNN is gives more accuracy when know the K- value and it is more efficient when the distance measure value is known. Finally, the Deep Learning algorithms are CNN and RNN gives more accuracy comparing with other algorithms for huge and multiclass text documents. For multi class classification, CNN gives best results and more efficient. Evaluation measures used for finding the accuracy and recall for identifying the best classifier mode for high accuracy and less time consumption.

References

- [1] Aizhang, Guo, and Yang Tao. 2016. "Based on Rough Sets and the Associated Analysis of KNN Text Classification Research." Proceedings - 14th International Symposium on Distributed Computing and Applications for Business, Engineering and Science, DCABES2015 (3): 485–88.
- [2] Baharudin, Baharum, Lam Hong Lee, and Khairullah Khan. 2010. "A Review of Machine Learning Algorithms for Text-Documents Classification." Journal of Advances in Information Technology 1(1).
- [3] Bhavani, A., Dr. Nageswara Rao Moparthy, "SPEECH RECOGNITION USING THE NN." 11(12) 2020: 2663–71.
- [4] Cai, Jingjing, Jianping Li, Wei Li, and Ji Wang. 2019. "Deep learning Model Used in Text Classification." 2018 15th International Computer Conference on Wavelet Active Media Technology and Information Processing, ICCWAMTIP 2018: 123–26.
- [5] Dhar, Ankita, Niladri Sekhar Dash, and Kaushik Roy. 2018. "Classification of Text Documents through Distance Measurement: An Experiment with Multi-Domain Bangla Text Documents." Proceedings - 2017 3rd International Conference on Advances in Computing, Communication and Automation (Fall), ICACCA 2017 2018-Janua: 1–6.
- [6] Ikonomakis, M., Sotos Kotsiantis, and V. Tampakas. 2005. "Text Classification Using Machine Learning Techniques." WSEAS Transactions on Computers 4(8): 966–74.

- [7] Jing, Ouyang. 2020. "Research on English Text Information Filtering Algorithm Based on SVM." *Proceedings of 2020 IEEE International Conference on Power, Intelligent Computing and Systems, ICPICS 2020*: 1001–4.
- [8] Kolluri, Johnson, and Shaik Razia. 2020. "Text Classification Using Naïve Bayes Classifier." *Materials Today: Proceedings (xxxx)*. <https://doi.org/10.1016/j.matpr.2020.10.058>.
- [9] Li, Zhonghao. 2019. "A Classification Retrieval Approach for English Legal Texts." *Proceedings - 2019 International Conference on Intelligent Transportation, Big Data and Smart City, ICITBS 2019*: 220–23.
- [10] Liu, Cai Zhi, Yan Xiu Sheng, Zhi Qiang Wei, and Yong Quan Yang. 2018. "Research of Text Classification Based on Improved TF-IDF Algorithm." *2018 IEEE International Conference of Intelligent Robotic and Control Engineering, IRCE 2018 (2)*: 69–73.
- [11] Liu, Kan, and Lu Chen. 2019. "Medical Social Media Text Classification Integrating Consumer Health Terminology." *IEEE Access* 7: 78185–93.
- [12] Lu, Hongbiao, Xiaobao Liu, Yanchao Yin, and Zhicheng Chen. 2019. "A Patent Text Classification Model Based on Multivariate Neural Network Fusion." *2019 6th International Conference on Soft Computing and Machine Intelligence, ISCMi 2019*: 61–65.
- [13] Ma, Houfeng, Xinghua Fan, and Ji Chen. 2008. "An Incremental Chinese Text Classification Algorithm Based on Quick Clustering." *Proceedings - International Symposium on Information Processing, ISIP 2008 and International Pacific Workshop on Web Mining and Web-Based Application, WMA 2008*: 308–12.
- [14] Ma, Yajing, Yonghong Li, Xiaolong Wu, and Xiang Zhang. 2018. "Chinese Text Classification Review." *Proceedings - 9th International Conference on Information Technology in Medicine and Education, ITME 2018*: 737–39.
- [15] Malakar, Susanta, and Werapon Chiracharit. 2020. "Thai Text Detection and Classification Using Convolutional Neural Network." *2020 59th Annual Conference of the Society of Instrument and Control Engineers of Japan, SICE 2020 (September)*: 99–102.
- [16] Mironczuk, Marcin Michał, and Jarosław Protasiewicz. 2018. "A Recent Overview of the State-of-the-Art Elements of Text Classification." *Expert Systems with Applications* 106: 36–54.
- [17] Myaeng, Sung Hyon, Kyoung Soo Han, and Hae Chang Rim. 2006. "Some Effective Techniques for Naive Bayes Text Classification." *IEEE Transactions on Knowledge and Data Engineering* 18(11): 1457–66.
- [18] Utomo, Muhammad Romi Ario, and Yuliant Sibaroni. 2019. "Text Classification of British English and American English Using Support Vector Machine." *2019 7th International Conference on Information and Communication Technology, ICoICT 2019*: 1–6.
- [19] Venkatesh, and K. V. Ranjitha. 2019. "Classification and Optimization Scheme for Text Data Using Machine Learning Naïve Bayes Classifier." *2018 IEEE World Symposium on Communication Engineering, WSCE 2018*: 33–36.
- [20] Y. Zheng, "An exploration on text classification with classical machine learning algorithm," *Proc. - 2019 Int. Conf. Mach. Learn. Big Data Bus. Intell. MLBDDBI 2019*, pp. 81–85, 2019, doi: 10.1109/MLBDDBI48998.2019.00023.